

- 6 (a) (i) 1. face: PQRS, polarity: positive
face: JKLM, polarity: negative [1 for both correct faces
or 1 for corresponding face & polarity correct]
2. Using Fleming's Left Hand Rule, force on electrons is downward / electrons will accumulate on the face JKLM. [1]
- (ii) (As charges separates,) an electric field is created between PQRS and JKLM [1]
- Maximum value is reached when electric force on electron is equal and opposite to magnetic force on electron. [1]
- (iii) Using $I = nAvq$
 $6.3 \times 10^{-4} = (1.3 \times 10^{29})(d)(0.10 \times 10^{-3})(E / B)(1.6 \times 10^{-19})$ [1 for $v = E/B$]
 $3.0288 \times 10^{-10} = (d)(E)(1 / B)$
 $3.0288 \times 10^{-10} = (d)(\Delta V / d)(1 / B)$ [1 for $E = \Delta V/d$]
 $\Delta V = (3.0288 \times 10^{-10})(B)$
 $= (3.0288 \times 10^{-10})(4.6 \times 10^{-3})$
 $= 1.4 \times 10^{-12} \text{ V}$ [1]
- (b) (i) Light damping [1]
- (ii) Oscillating sheets cuts magnetic flux and hence by Faraday's law, an emf is induced. [1]
- Induced emf causes eddy currents to flow in sheets. [1]
- By Lenz's law,
either currents in sheets cause resistive force to oppose motion
or currents in sheets dissipate heat energy resulting in loss in KE [1]
- Smaller currents in Y or larger currents in X, so dashed line is X. [1]
- (c) (i) Max emf = $NBA\omega = (500)(5.0 \times 10^{-2})(2.5 \times 10^{-2})(2\pi \times 50) = 196 \text{ V}$ [1]
r.m.s. emf = $196 / (\sqrt{2})$ [1]
= 140 V (Shown)
- (ii) max emf across secondary coil = $(50 / 2000)(196) = 4.9 \text{ V}$ [1]
- (iii) 1. The supply voltage is less than emf of cell (1.2 V) **or** less than voltage needed across diodes (1.4 V) [1]
- less than 2.6 V, the sum of emf and voltage need across diodes [1]
2. $3.75 - 4.35 \times 10^{-4} \text{ C}$ [1]
OR
 $3.90 - 4.20 \times 10^{-4} \text{ C}$ [2]